

The exponential model describes population growth in an idealized, unlimited environment

Populations of all species have the potential to expand greatly when resources are abundant. To appreciate the potential for population increase, consider a bacterium that can reproduce by fission every 20 minutes under ideal laboratory conditions. There would be two bacteria after 20 minutes, four after 40 minutes, and eight after 60 minutes. If reproduction continued at this rate for a day and a half without mortality, there would be enough bacteria to form a layer 30 cm deep over the entire globe! But unlimited growth does not occur for long in nature, where individuals typically have access to fewer resources as a population grows. Nonetheless, ecologists study population growth in ideal, unlimited environments to reveal how fast populations are capable of growing and the conditions under which rapid growth might actually occur.

Changes in Population Size

Imagine a population consisting of a few individuals living in an ideal, unlimited environment. Under these conditions, there are no external limits on the abilities of individuals to harvest energy, grow, and reproduce. The population will increase in size with every birth and with the immigration of individuals from other populations, and it will decrease in size with every death and with the emigration of individuals out of the population. We can thus define a change in population size during a fixed time interval with the following verbal equation:

$$\begin{array}{rcccl} \text{Change in} & & \text{Immigrants} & & \text{Emigrants} \\ \text{population} & = & \text{entering} & - & \text{leaving} \\ \text{size} & & \text{population} & & \text{population} \end{array}$$

For now, we will simplify our discussion by ignoring the effects of immigration and emigration.

We can use mathematical notation to express this simplified relationship more concisely. If N represents population size and t represents time, then ΔN is the change in population size and Δt is the time interval (appropriate to the life span or generation time of the species) over which we are evaluating population growth. (The Greek letter delta, Δ , indicates change, such as change in time.) Using B for the number of births in the population during the time interval and D for the number of deaths, we can rewrite the verbal equation:

$$\frac{\Delta N}{\Delta t} = B - D$$

Population ecologists often are most interested in changes in population size—the number of individuals that are added to or subtracted from a population during a given time interval, symbolized by R . Here, R represents the *difference* between the number of births (B) and the number of deaths (D) that

occur in the time interval. Thus, $R = B - D$, and we can simplify our equation by writing the following:

$$\frac{\Delta N}{\Delta t} = R$$

Next, we can convert our model to one in which changes in population size are expressed on a per individual (per capita) basis. The *per capita* change in population size ($r_{\Delta t}$) represents the contribution that an average member of the population makes to the number of individuals added to or subtracted from the population during the time interval Δt . If, for example, a population of 1,000 individuals increases by 16 individuals per year, then on a per capita basis, the annual change in population size is $16/1,000$, or 0.016. If we know the annual per capita change in population size, we can use the formula $R = r_{\Delta t}N$ to calculate how many individuals will be added to (or subtracted from) a population each year. For example, if $r_{\Delta t}$ is 0.016 and the population size is 500,

$$R = r_{\Delta t}N = 0.016 \times 500 = 8 \text{ per year}$$

Since the number of individuals added to (or subtracted from) the population (R) can be expressed on a per capita basis as $R = r_{\Delta t}N$, we can revise our population growth equation to take this into account:

$$\frac{\Delta N}{\Delta t} = r_{\Delta t}N$$

Remember that our equation is for a specific time interval (often one year). However, many ecologists prefer to use differential calculus to express population growth as a rate of change *at each instant in time*:

$$\frac{dN}{dt} = rN$$

In this case, r represents the per capita change in population size that occurs at each instant in time (whereas $r_{\Delta t}$ represented the per capita change that occurred during the time interval Δt). If you have not yet studied calculus, don't be intimidated by the last equation; it is similar to the previous one, except that the time intervals Δt are very short and are expressed in the equation as dt . In fact, as Δt becomes shorter, $r_{\Delta t}$ and r become increasingly close to one another in value.

Exponential Growth

Earlier we described a population whose members all have access to abundant food and can reproduce at their physiological capacity. In some cases, a population that experiences such ideal conditions increases in size by a constant proportion at each instant in time. When this occurs, the pattern of growth that results is called **exponential population growth**. The equation for exponential growth is the one presented at the end of the previous section, namely:

$$\frac{dN}{dt} = rN$$